

Bellabeat Product Analysis Case Study

How Can a Wellness Company Play It Smart?

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1 Introduction

This case study examines how consumers use smart fitness devices and how those usage patterns could support Bellabeat's marketing strategy. Bellabeat is a health-focused technology company that develops smart wellness products for women. The analysis will use publicly available Fitbit fitness-tracker data to identify trends in activity, sleep, and device usage. The findings will then be applied to one Bellabeat product to provide practical marketing recommendations.

2 Ask

2.1 Business Task

The business task is to analyse smart-device usage data to identify patterns in consumer activity, sleep, and device use. The findings will be applied to the Bellabeat app to help the company develop marketing strategies that encourage users to engage more consistently with their wellness data and Bellabeat products.

2.2 Business Problem

Bellabeat wants to strengthen its position in the global smart-device market. To support this goal, the company needs a better understanding of how consumers currently use fitness trackers and how those behaviours could inform product features, communication, and marketing campaigns.

2.3 Main Business Questions

- What trends can be identified in smart-device usage?
- How could these trends apply to Bellabeat customers?
- How could the findings influence Bellabeat's marketing strategy?

2.4 Selected Bellabeat Product

This case study will focus on the Bellabeat app. The app is suitable because it brings together information on activity, sleep, stress, menstrual health, and mindfulness. Insights from general fitness-tracker behaviour can therefore be applied to how Bellabeat communicates the value of the app and encourages regular engagement.

2.5 Key Stakeholders

- Urška Sršen, Cofounder and Chief Creative Officer
- Sando Mur, Cofounder and executive team member
- Bellabeat marketing analytics team
- Bellabeat executive team
- Current and potential Bellabeat customers

3 Prepare

3.1 Data Source

The analysis will use the publicly available Fitbit Fitness Tracker Data dataset provided through Kaggle. The dataset description states that the data was collected from 30 eligible Fitbit users. However, the daily activity file used in this analysis contained 33 unique user IDs, while the sleep file contained 24 unique users. The dataset is suitable for identifying general trends in how consumers use smart fitness devices.

3.2 Data Organisation

The dataset contains several CSV files organised by type of measurement. These include daily activity, sleep, heart rate, steps, calories, and minute-level activity records. For this analysis, the main files will be selected based on their relevance to the business questions.

3.3 Files Selected for Analysis

- dailyActivity_merged.csv
- sleepDay_merged.csv

3.4 Data Credibility and Limitations

The dataset is relevant because it contains real smart-device usage data. However, it has several limitations. The dataset description refers to 30 eligible Fitbit users, although the daily activity file contained 33 unique user IDs. In either case, the sample is too small to represent all smart-device consumers. The demographic characteristics of the users are not provided, and the dataset may not specifically represent Bellabeat's target audience of women. The data also covers a limited period, so the findings should be interpreted as exploratory rather than fully representative.

3.5 Privacy and Licensing

The dataset is publicly available and does not include personally identifiable information. The users consented to share their fitness-tracker data. The data is available under a CC0 public-domain licence, which allows it to be used for analysis.

3.6 Relevance to the Business Question

The dataset can help identify patterns in physical activity, steps, calories, sedentary behaviour, and sleep. These trends can be used to understand how consumers engage with smart devices and how Bellabeat could encourage more consistent use of its app and wellness products.

4 Process

4.1 Tools Used

Python was used in Google Colab to import, clean, transform, combine, and analyse the Fitbit data. The pandas library was used because it made it possible to work efficiently with multiple CSV files, check data quality, calculate new variables, and produce summary statistics.

4.2 Data Cleaning

The dailyActivity_merged.csv file contained 940 rows and 15 columns, while the sleepDay_merged.csv file contained 413 rows and 5 columns.

The ActivityDate and SleepDay columns were converted from text into date format. No invalid date values were identified.

The daily activity dataset contained no duplicate rows. The sleep dataset contained three duplicate records, which were removed, leaving 410 sleep records.

Both datasets were checked for missing values. No missing values were identified.

4.3 User Coverage

The daily activity dataset contained records from 33 unique users. The sleep dataset contained records from 24 unique users. This means that sleep-related findings are based on fewer participants than the activity analysis.

4.4 New Variables Created

The following variables were created to support the analysis:

- **DayOfWeek:** The day on which each activity or sleep record occurred.
- **TotalActiveMinutes:** The combined total of very active, fairly active, and lightly active minutes.
- **SleepHours:** Total minutes asleep converted into hours.
- **TimeInBedHours:** Total time in bed converted into hours.
- **SleepEfficiency:** The percentage of time in bed that was spent asleep.
- **ActivityLevel:** A classification based on each user's average daily steps.

4.5 Data Integration

The cleaned daily activity and sleep datasets were combined using user ID and date. The merged dataset contained 410 matched records from 24 users.

4.6 Final Cleaned Files

Three cleaned files were exported and stored separately from the original data: a cleaned daily activity dataset, a cleaned sleep dataset, and a merged activity-and-sleep dataset.

5 Analyze

5.1 Daily Activity Overview

Users recorded an average of 7,637.91 steps per day and recorded an average distance of 5.49. They spent an average of 21.16 minutes in very active activity, 13.56 minutes in fairly active activity, and 192.81 minutes in lightly active activity each day.

Average total active time was 227.54 minutes per day. However, users also recorded an average of 991.21 sedentary minutes per day, equivalent to approximately 16.5 hours.

5.2 Calories and Steps

The correlation between total steps and calories burned was 0.59, indicating a moderate positive relationship. In general, users who took more steps also burned more calories.

This relationship does not prove that steps alone caused the increase in calories, because other factors such as activity intensity, body size, and unrecorded exercise may also influence calorie expenditure.

5.3 Sleep Patterns

Users slept an average of 6.99 hours per night and spent an average of 7.64 hours in bed. Average sleep efficiency was 91.65%, meaning that most of the time spent in bed was spent asleep.

Sleep findings were based on 24 users rather than all 33 activity users, because not every participant recorded sleep data.

5.4 Relationship Between Activity and Sleep

The relationship between total steps and sleep duration was weak, with a correlation of -0.19. The relationship between total active minutes and sleep duration was also very weak, with a correlation of -0.07.

Sedentary minutes had a stronger relationship with sleep duration. The correlation was -0.60, suggesting that users with more sedentary time tended to sleep fewer hours. This is an association only and does not prove that sedentary behaviour caused reduced sleep.

5.5 Activity by Day of the Week

Saturday recorded the highest average number of steps, with approximately 8,153 steps, followed closely by Tuesday with about 8,125 steps. Sunday recorded the lowest average activity level at approximately 6,933 steps.

Saturday also had the highest average active time, while Sunday had the lowest. These results suggest that user activity varied across the week, with stronger activity on Saturday and Tuesday and weaker activity on Sunday.

5.6 Sleep by Day of the Week

Users recorded their longest average sleep on Sunday at 7.55 hours. Wednesday followed with an average of 7.24 hours. Thursday recorded the shortest average sleep duration at 6.69 hours.

Sunday had the lowest average sleep efficiency at 90.51%, while Thursday had the highest at 92.27%. The differences in sleep efficiency were relatively small.

5.7 User Activity Levels

Users were classified according to their average daily steps. Lightly active and fairly active users each represented 27.27% of the sample. Sedentary users represented 24.24%, while very active users represented 21.21%.

Overall, 51.51% of users were either sedentary or lightly active. This suggests that many smart-device users may benefit more from achievable goals, gentle progress tracking, and regular movement reminders than from highly demanding fitness targets.

6 Share

6.1 Visualisations

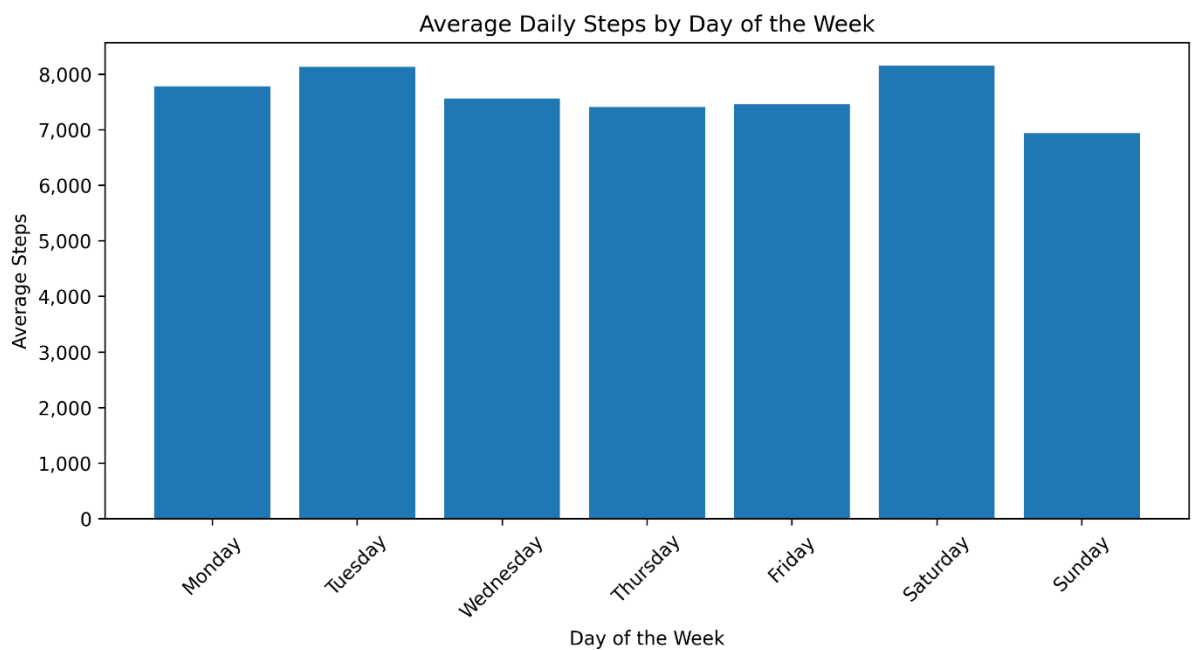


Figure 1: Average Daily Steps by Day of the Week

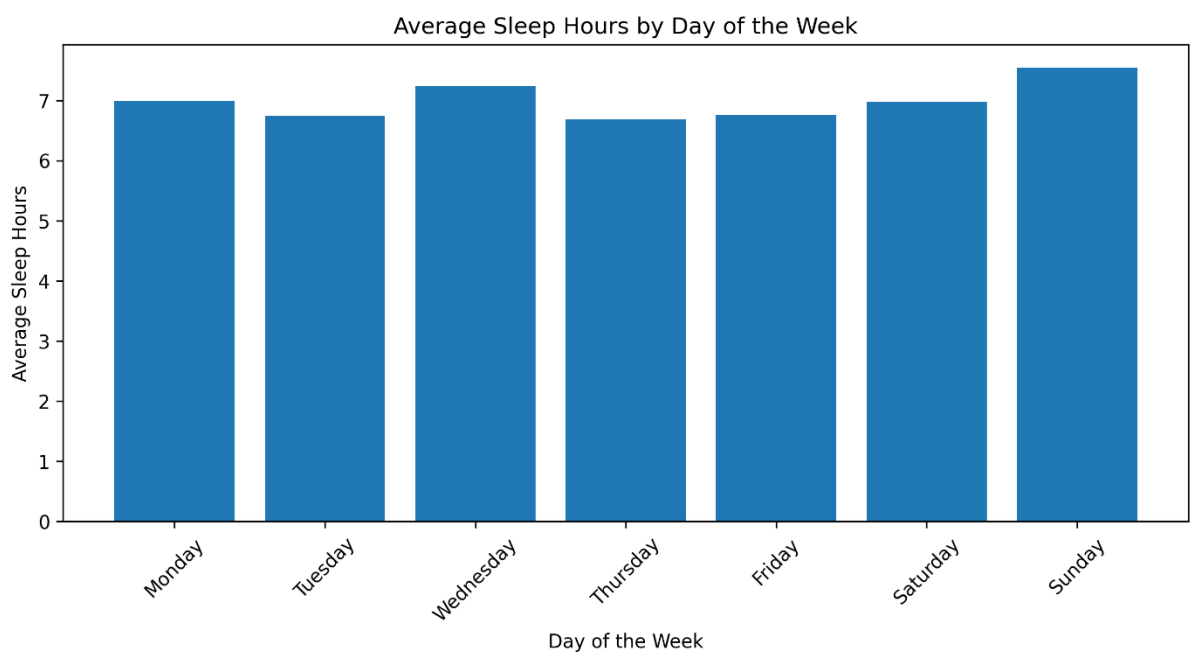


Figure 2: Average Sleep Hours by Day of the Week

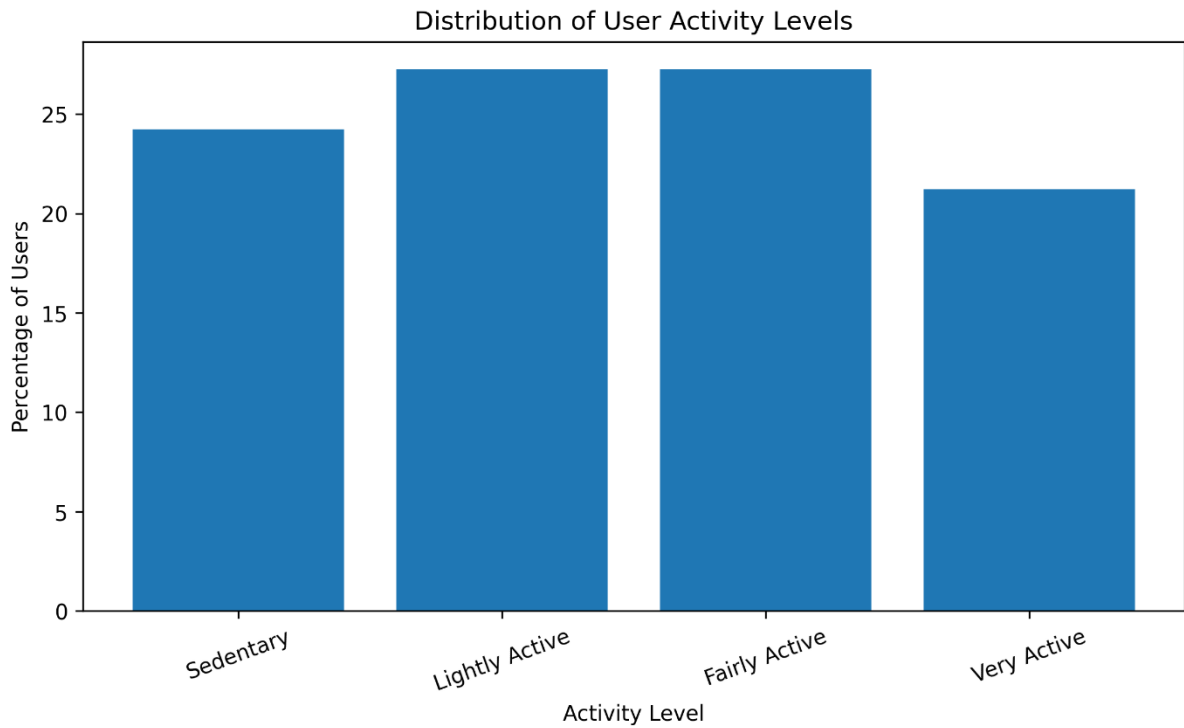


Figure 3: Distribution of User Activity Levels



Figure 4: Relationship Between Daily Steps and Calories Burned

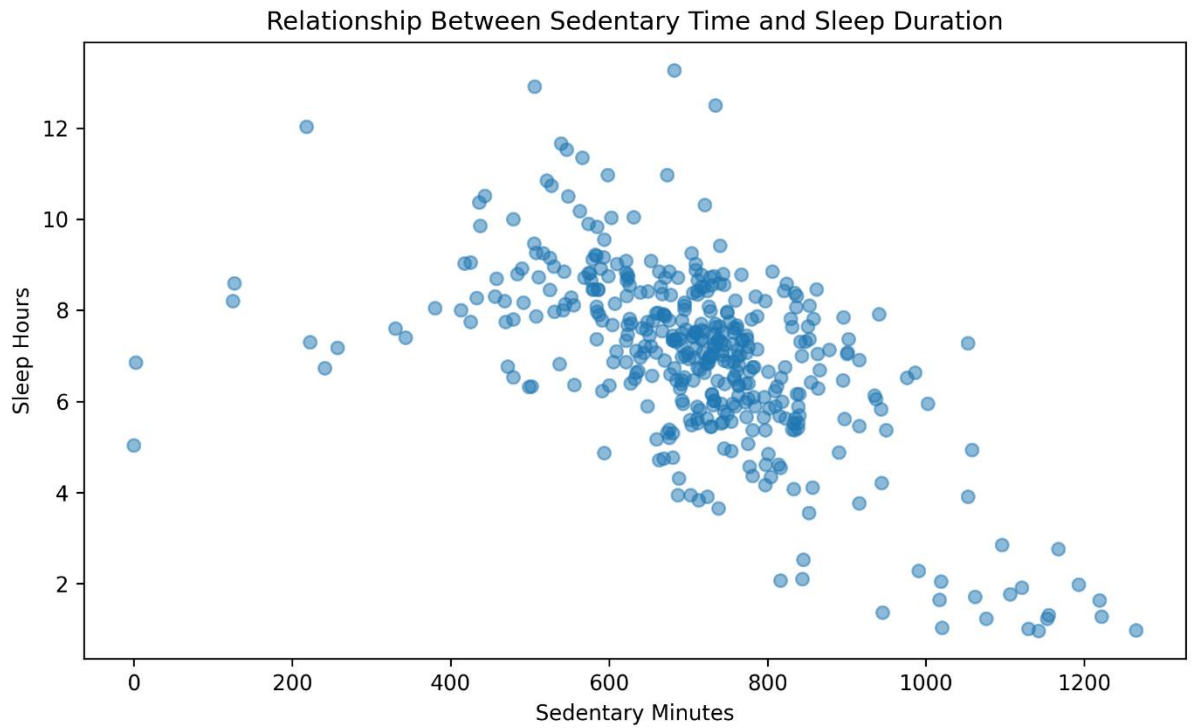


Figure 5: Relationship Between Sedentary Time and Sleep Duration

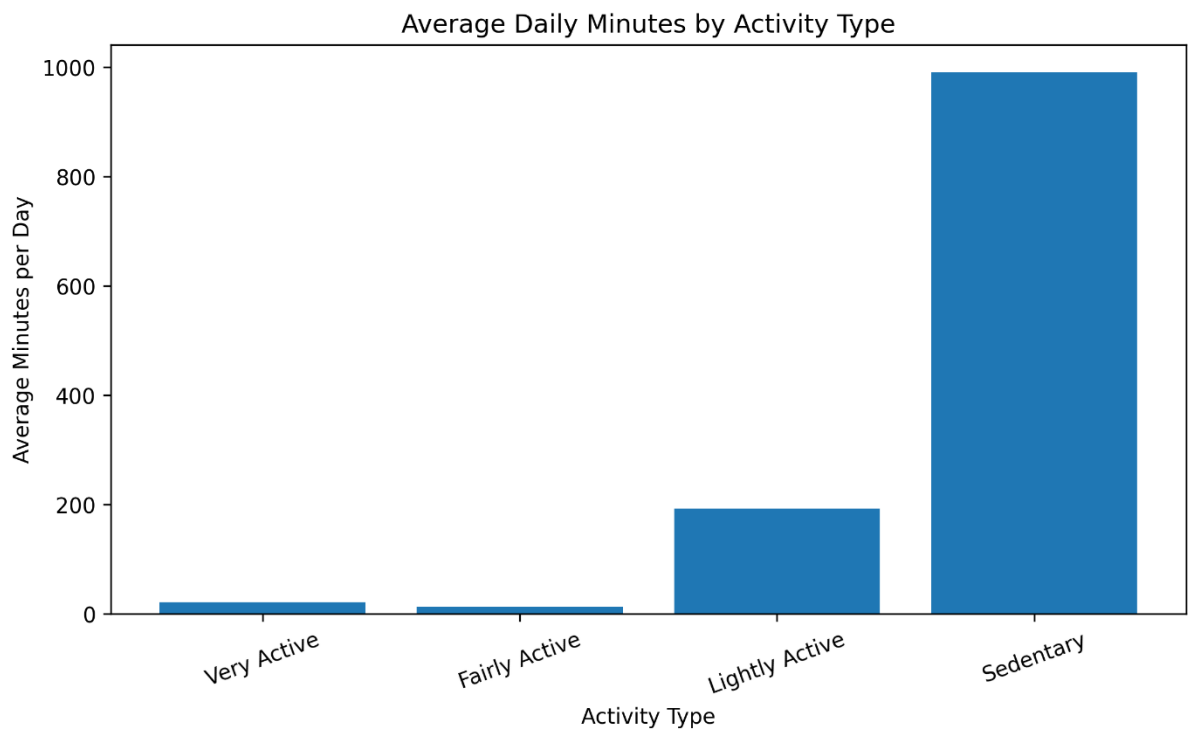


Figure 6: Average Daily Minutes by Activity Type

6.2 Key Findings

The visualisations show that users recorded relatively modest activity levels overall. Average daily steps remained below 10,000 throughout the week, and more than

half of the users were either sedentary or lightly active. Sedentary time was also substantially higher than time spent in active movement.

Users slept an average of approximately seven hours per night. Sleep duration varied slightly across the week, with the longest average sleep recorded on Sunday and the shortest on Thursday.

A moderate positive relationship was identified between daily steps and calories burned, suggesting that users who moved more generally burned more calories. A moderate negative relationship was also identified between sedentary time and sleep duration, meaning that users with more sedentary minutes tended to sleep fewer hours.

6.3 Main Answer to the Business Questions

The data suggests that many smart-device users are not highly active and may benefit from simple, achievable wellness goals rather than intense fitness targets. Users also show meaningful interest in both activity and sleep tracking, making it important for Bellabeat to present wellness as a combination of movement, rest, and daily habits.

These insights can be applied to the Bellabeat app by promoting personalised movement reminders, sleep guidance, achievable step goals, and progress-based encouragement. Marketing should position the app as a supportive wellness companion for everyday users, not only as a tool for highly active consumers.

7 Act

7.1 Conclusion

The analysis identified several useful trends in smart-device usage. Users averaged fewer than 10,000 steps per day, spent a large amount of time in sedentary activity, and slept approximately seven hours per night. More than half of the users were classified as sedentary or lightly active.

The findings also showed a moderate positive relationship between steps and calories burned, as well as a moderate negative relationship between sedentary time and sleep duration. These patterns suggest that Bellabeat should focus on helping users build healthier daily routines through manageable activity goals, movement reminders, and sleep support.

7.2 Recommendations

Recommendation 1: Promote achievable daily goals

Bellabeat should use the app to set realistic and personalised step and activity goals based on each user's current behaviour. Progress messages, small milestones, and weekly challenges could help sedentary and lightly active users improve gradually without feeling overwhelmed.

Recommendation 2: Reduce sedentary behaviour

The app should provide timely movement reminders when users have been inactive for long periods. Bellabeat could also introduce short activity suggestions, such as walking breaks, stretching, or brief wellness challenges that fit easily into a user's daily routine.

Recommendation 3: Connect activity and sleep guidance

Bellabeat should position the app as a complete wellness tool by combining activity and sleep insights. The app could provide personalised summaries showing how movement, sedentary time, and sleep patterns may relate to one another, while encouraging users to develop balanced routines.

7.3 Marketing Application

Bellabeat's marketing should focus on everyday wellness rather than only intense exercise. Campaigns could present the Bellabeat app as a supportive companion that helps users move more, rest better, and understand their habits.

Digital marketing content could include simple movement challenges, sleep-improvement tips, personalised progress stories, and reminders that small changes can lead to healthier routines.

7.4 Suggested Next Steps

Bellabeat should test these recommendations with a small group of users and measure app engagement, completion of activity goals, response to movement reminders, and changes in sleep-tracking behaviour.

Future analysis should use a larger and more diverse sample, include demographic information, and collect data over a longer period. Bellabeat-specific customer data would also provide more reliable insights than the general Fitbit dataset.

8 References

Google Data Analytics Professional Certificate. (2026). *Case Study 2: How Can a Wellness Company Play It Smart?*

Mobius. (2016). *FitBit Fitness Tracker Data*. Kaggle.

Bellabeat. *Bellabeat Products and Wellness App Information*.